Jhonathan Prieto Rojas, Ph.D.

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Links: Google Scholar | Personal website | LinkedIn
Current location: Bogotá, Colombia

ACADEMIC PROFILE

Electrical engineering academic with more than ten years of international experience in research-led teaching, research supervision, and academic leadership. Research spans flexible and stretchable semiconductor devices, thermoelectric energy harvesting, wearable and distributed sensing systems, and emerging work in AI-assisted PCB design automation. This work combines micro/nanofabrication, sensor integration, low-power electronics, wireless communication, and electrical, thermal, and mechanical characterisation for health, geophysical, and renewable-energy applications. Track record includes 37 peer-reviewed journal papers, four granted U.S. patents, one published U.S. patent application, and more than USD 85,000 secured as Principal Investigator across four projects. International academic and industrial experience spans Colombia, Saudi Arabia, the United States, and Germany. Experienced in curriculum and course development, postgraduate and capstone supervision, multi-section coordination, ABET accreditation, and academic governance.

Publications: 37 peer-reviewed journal papers — **U.S. patents:** 4
Total citations: 1580+ (Scopus), 2100+ (Google Scholar) — ***h*-index:** 20 (Scopus), 24 (Google Scholar)

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Professor | Sep. 2015 – Aug. 2026
King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Led interdisciplinary research on wearable and distributed sensing systems incorporating energy harvesting, sensor-node development, and system validation for health, geophysical, and renewable-energy applications.
- Secured more than USD 85,000 as Principal Investigator across four projects and contributed as Co-Investigator to three additional multi-institutional projects in geophysical sensing, optical health monitoring, and crowd-management systems.
- Developed and taught undergraduate and postgraduate courses in circuits, electronics, digital systems, semiconductor devices, and micro/nanofabrication using research-led teaching and active, project-based learning.
- Supervised two completed MSc theses, two ongoing PhD dissertations, now as external co-supervisor, and ten capstone projects, several of which produced peer-reviewed publications and award-winning prototypes.
- Provided academic leadership as Electronics Group Coordinator; led the development of an interdisciplinary concentration in IC Design and Fabrication; and supported program review, ABET accreditation, quality assurance, graduate admissions, and faculty recruitment.

Research Scholar — Concurrent appointment | May 2021 – Aug. 2026
Interdisciplinary Research Center for Advanced Materials, KFUPM, Dhahran, Saudi Arabia

- Led funded research projects as Principal investigator in thermoelectric energy harvesting for wearable applications, and dust mitigation technologies.

Postdoctoral Fellow | Oct. 2014 – Aug. 2015
King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Designed and fabricated flexible and stretchable Bi_2Te_3/Sb_2Te_3 thermoelectric generators using helical, foldable, and paper-based architectures to improve mechanical compliance and temperature-gradient control.
- Developed low-cost microfabrication approaches and evaluated their electrical, thermal, and mechanical performance.

Visiting Researcher | May 2015 – Jun. 2015
University of Illinois at Urbana-Champaign, USA
Co-advisor: Prof. John Rogers

- Adapted transfer printing technology to improve the manufacturability of high-performance flexible electronics and enable 3D integration using flexible Si (100) fabrics.

Graduate Research Assistant | Sep. 2009 – Jun. 2014
Integrated Nanotechnology Lab, KAUST, Thuwal, Saudi Arabia

- Developed CMOS-compatible micro/nanofabrication processes for transforming rigid silicon wafers into flexible, high-performance electronic platforms for More-than-Moore applications.
- Fabricated and characterised MOS capacitors, MIM capacitors, and MOSFETs to validate retained device performance, contributing to publications and patents on flexible silicon electronics and self-powered microsystems..

Visiting Researcher, *KAUST Global Collaborative Research (GCR) Program* | Jun. 2010 – Aug. 2010
The Pennsylvania State University, State College, USA
Co-advisor: Prof. Bruce Logan

- Designed and successful fabricated a micro-sized CNT-based Microbial Fuel Cell (MFC) in collaboration with the Materials Research Institute and Environmental Engineering Department.

Intern, "Product-to-Market" Department | Mar. 2008 – Jul. 2008
Micronas GmbH, Munich, Germany

- Evaluated, diagnosed, and corrected functionality issues in the hardware and software of an MPEG-2 video converter board mounted on a Spartan3 FPGA.
- Delivered a fully functional board with technical documentation after a five-month internship.

EDUCATION

King Abdullah University of Science and Technology, Saudi Arabia | Jan. 2011 – Jun. 2014
PhD Electrical Engineering, Electrophysics track (GPA: 3.84/4.0)
Dissertation Title: "Towards Flexible Self-Powered Micro-scale Integrated Systems"
Academic advisor: Prof. Muhammad M. Hussain, KAUST.
External PhD committee member: Prof. John Rogers, UIUC, USA.

King Abdullah University of Science and Technology, Saudi Arabia | Sep. 2009 – Dec. 2010
MSc Electrical Engineering, Solid State Devices track (GPA: 3.93/4.0)
Thesis Title: "Advanced Nanofabrication Process Development for Self-Powered SoC"
Academic advisor: Prof. Muhammad M. Hussain

Technical University of Munich, Munich, Germany | Oct. 2007 – Mar. 2008
Exchange Student, Electronics Engineering (Focus on VLSI design, hardware programming and Nanotechnology)

National University of Colombia, Colombia | Jan. 2003 – Mar. 2009
BSc in Electronics Engineering (GPA: 4.3/5.0, Ranked Top 3 in Engineering College)
Thesis Title: "Study and Optimization of a Development Card for CPLD XC9572XL"

TEACHING AND ACADEMIC EXPERIENCE

My teaching experience includes contributing to the development of three academic programs, leading one of them, and creating and teaching three specialised senior-level and postgraduate courses in micro/nanoelectronics, semiconductor device manufacturing, and micro/nanotechnology and biosensors. I have also coordinated multi-section courses and laboratories in circuits, electronics, and digital logic. My teaching combines active and project-based learning, team problem-solving, interactive polling, laboratory demonstrations, and research-linked capstone projects. I have adapted laboratories for online delivery and used Blackboard Ultra, Mentimeter, Gradescope, and other digitally enabled approaches. These contributions were recognised with the KFUPM Excellence in Teaching Award for 2020–2021.

Academic Programs Developed:

- Academic concentration (CX) in IC Design and Fabrication (EE, Phys, COE) (*Main Coordinator*)
- Academic concentration (CX) in Bioelectronics and Sensors (EE, Bio, COE) (*Team member*)
- Academic concentration (CX) in Mechatronics (ME, EE, COE) (*Team member*)

** Each CX program is a four-course interdisciplinary concentration for senior students.*

Courses Developed & Taught:

- EE436 – Introduction to Micro & Nanoelectronics
- EE439 – Micro/Nanotechnology & Biosensors
- EE649 – Solid State Devices & Manufacturing (Graduate level)

Courses & Laboratories Taught:

- EE200 – Digital Logic Circuit Design (Multi-section Coordinator)
- EE201 – Electric Circuits I (Multi-section Coordinator)
- EE203 – Electronics I (Multi-section Coordinator)
- EE203L – Electronics I Lab (Multi-section Coordinator)
- EE204 – Fundamentals of Electrical Circuits (Non-EE Students)
- EE234 – Electronics & Microcontrollers (Non-EE Students) (Multi-section Coordinator)
- EE236 – Electronic Circuits (Non-EE Students)
- EE303 – Electronics II (Multi-section Coordinator)
- EE304L – Electronics II Lab (Multi-section Coordinator)

RESEARCH GRANTS

Projects as Principal Investigator (Total funding secured > USD 85,000):

1. *Interdisciplinary Research Center Grant (~USD 20,000)* **| Feb. 2023 – Dec. 2024**
   Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase II]
2. *Interdisciplinary Research Center Grant (~USD 13,500)* **| Jul. 2021 – Dec. 2022**
   Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase I]
3. *KFUPM Internal Funded Grant (~USD 39,000)* **| Apr. 2017 – Jun. 2018**
   Advanced Study of Microfabricated Stretchable Silicon-based Structures
4. *KFUPM Junior Faculty Research Grant (~USD 12,800)* **| Feb. 2016 – Oct. 2016**
   Optimization of Spiral-based Structures for Stretchable Electronics

Projects as Co-Investigator:

1. *Georgia Tech-KFUPM Collaboration Grant* **| May 2019 – Nov. 2021**
   Design and Implementation of a Wireless Geophone Network for Seismic Data Collection
2. *KACST Technology Innovation Center - EE2381* **| Jan. 2018 – Dec. 2020**
   Solid-State Lighting: Optical-based Sensing for Advanced Health Monitoring System
3. *KAUST-KFUPM Collaboration Grant* **| Oct. 2016 – Oct. 2019**
   Biometric data-based crowd management and risk mitigation for high population event

RESEARCH SUPERVISION

Primary supervisor of two completed MSc theses and two ongoing PhD dissertations (now as external co-supervisor); committee member for two completed MSc theses; supervisor of ten senior-design projects and more than twelve industrial internships.

Undergraduate:

- Advisor for 10 undergraduate senior design projects related to dust mitigation technologies, electronic sensing systems for geological exploration and wearable applications.
- Advisor for over 12 undergraduate students in their Industrial Internships and Summer Training experiences.
- Academic advisor to approximately 20 undergraduate students per semester, guiding degree planning and academic performance.

Graduate:

M.Sc. Thesis Supervisor:

- Title: Flow-Field-Based Routing Algorithms for Efficient PCB Design Automation. (Completed, 2026)
- Title: Optimization of Spiral-based Structures for Stretchable Electronics. (Completed, 2017)

M.Sc. Committee Member:

- Title: Dynamics of MDOF Metamaterial Resonators Attached to a Periodic Waveguide. (Completed, 2022)
- Title: Probabilistic Sampling Towards Tunable Autonomous Sensing. (Completed, 2022)

PhD Dissertations as External Co-supervisor:

- Title: Hybrid Dust Mitigation Techniques for Solar Panels in Arid and Semi-Arid Regions. (In progress, Proposal defended, Expected graduation: June 2027)
- Title: Development of Kirigami-inspired Thermoelectric Generators for Self-powered Sensing Applications. (In progress, Proposal defended, Expected graduation: Dec. 2026)

SELECTED JOURNAL PUBLICATIONS (Total: 37, 15 as first author, 6 as corresponding author, 7 cover pages)

1. S. S. Ali, S. Firdous, S. Dieng, **J. P. Rojas**, T. A. Saleh, "MXene-assisted Bi_2Te_3 pellet-based thermoelectric generator," ***RSC Adv.*** (2026) DOI: 10.1039/D6RA06939H (Q2)
2. **J. P. Rojas,** "Design and evaluation of LET-based soft hinges for stress mitigation in flexible copper interconnects", ***Flex. Print. Electron.*** 10, 035016 (2025). (Q2)
3. A. A. Alduais, A. B. Javaid, M. A. Syed, **J. P. Rojas**, "Analysis of Electrodynamic Screens with Fractal Designs for Enhanced Performance", ***Arab. J. Sci. Eng.*** 50, 17575-17585 (2025). (Q2)
4. **J. P. Rojas,** R. Almazyad, A. AlHayyah, A. Alruhaiman, M. Almusharraf, S. Al-Dharrab, H. Attia, "Self-Powered End-to-End Wireless Sensor Network for Geophysical Explorations," ***IEEE Syst. J.*** 19, 107-118 (2025). (Q1)
5. H. S. Daraghma, D. B. Ferry, S. G. Rao, M. A. Hawwa, M. A. Gondal, **J. P. Rojas**, "Materials & Design Approaches for Enhanced Performance of Mechanically Compliant Thermoelectric Generators (TEGs): A Review", ***Smart Mater. Struct.*** 33, 103003 (2024). (Q2)
6. M. S. Kim, A. S. Almuslem, W. Babatain, R. R. Bahabry, U. K. Das, N. El-Atab, M. Ghoneim, A. M. Hussain, A. T. Kutbee, J. Nassar, N. Qaiser, **J. P. Rojas**, S. F. Shaikh, G. A. Torres Sevilla, M. M. Hussain, "Beyond Flexible: Unveiling the Next Era of Flexible Electronic Systems", **Adv. Mater.** 36, 2406424 (2024). (Q1) **{Top viewed Article 2025 in Adv. Mater.}**
7. **J.P. Rojas**, M.U. Rehman, A. Hussein, A. E'mar, Y. AlRaei, H. AlZaher, M. Mohandes, "Kirigami-Enabled Wearable Health and Crowd Monitoring System", ***Arab. J. Sci. Eng.*** 47, 3583-3595 (2022). (Q2)
8. M.U. Rehman, W. Babatain, S.F. Shaikh, D. Conchouso, N. Qaiser, M. M. Hussain, **J. P. Rojas,** "Stress concentration analysis and fabrication of silicon (100) based ultra-stretchable structures with parylene coating", ***Extreme Mech. Lett.*** 41, 101052 (2020). (Q1)
9. M.U. Rehman, **J.P. Rojas**, "Optimization of compound serpentine–spiral structure for ultra-stretchable electronics", ***Extreme Mech. Lett.*** 15, 44-50 (2017). (Q1)
10. **J.P. Rojas**, D. Conchouso, A. Arevalo, D. Singh, I. G. Foulds, M. M. Hussain, "Paper-based origami flexible and foldable thermoelectric nanogenerator", ***Nano Energy*** 31; 296-301 (2017). (Q1)
11. **J.P. Rojas**, D. Singh, D. Conchouso, A. Arevalo, I. G. Foulds, M. M. Hussain, "Stretchable helical architecture inorganic-organic hetero thermoelectric generator", ***Nano Energy*** 30; 691-699 (2016). (Q1)
12. **J. P. Rojas**, G. A. Torres Sevilla, S. B. Inayat, A. M. Hussain, M. T. Ghoneim, S. Ahmed, M. M. Hussain, "Transformational Silicon Electronics", ***ACS Nano*** 8 (2), 1468–1474 (2014). (Q1)
13. J. E. Mink§, **J. P. Rojas**§, B. E. Logan, M. M. Hussain, "Vertically Grown Multiwalled Carbon Nanotube Anode and Nickel Silicide Integrated High Performance Microsized (1.25 μL) Microbial Fuel Cell", ***Nano Letters*** 12 (2), 791–795 (2012). §*These authors contributed equally to this work.* (Q1)

SELECTED CONFERENCES AND PROCEEDINGS (Total: 35, 12 as first author, 1 Best Presentation Award)

1. **J. P. Rojas**, PV Dust Removal through Fractal-based Electrodynamic Screens. **KAUST Research Conference 2023: Sustainable Energy Materials and Technologies for a Low Carbon Future.** (Invited talk)(2023)
2. **J. P. Rojas**, A. Hussein, H. AlZaher and M. Mohandes, Stretchable Wearable Crowd Monitoring System**, 2021 4th Inter. Symp. on Adv. Elect. and Comm. Tech. (ISAECT)** (Paper Presentation) (2021)
3. A. Alduais, A. Javaid, M. Syed, M. Alhrtomi, **J. P. Rojas**, Fractal Based EDS Design for Increased Solar Panel Dust Removal Efficiency, **2021 MRS Fall Meeting & Exhibit**. (Paper Presentation) (2021)
4. **J. P. Rojas**, Origami and Kirigami stretchable electronics. **IEEE NEMS 2020** (Invited talk) (2020)
5. **J. P. Rojas**, Organic-inorganic heterostructures for stretchable electronics. **Proceedings of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications XI** 10982, 1098215 (Invited – Session Chair) (2019)
6. **J. P. Rojas**, M. U. Rehman, M. A. Albettar, D. Conchouso, A. Arevalo, D. Singh, I. Foulds, M. M. Hussain. "Folding and stretching a thermoelectric generator"; **Proceedings of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications X** 10639, 106390E (Invited – Session Chair) (2018)

U.S. PATENTS AND PATENT APPLICATIONS

1. Cylindrical-Shaped Nanotube Field Effect Transistor, M. M. Hussain, H. M. Fahad, C. E. Smith, **J. P. Rojas**. U.S. Patent US9224813B2.
2. Apparatus, system, and method for on-chip thermoelectricity generation, M. M. Hussain, H. M. Fahad, **J. P. Rojas**. U.S. Patent 9515245B2.

3. Integrated circuit manufacturing for low-profile and flexible devices, M. M. Hussain, **J. P. Rojas**. U.S. Patent US9209083B2.
4. Method for Producing Mechanically Flexible Silicon Substrate, M. M. Hussain, **J. P. Rojas**. U.S. Patent US9520293B2.
5. Flexible and foldable paper-substrate thermoelectric generator (TEG) **J. P. Rojas,** M. M. Hussain. U.S. Published Patent Application 20170244019A1

PROFESSIONAL SERVICE

Member, Assessment, Evaluation and Accreditation Committee | Jan. 2024 – Aug. 2026
Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Support the preparation of the Self-Study Report (SSR) for ABET accreditation.
- Coordinate the development and review of course specifications in the department.
- Assess the quality of teaching in the department.
- Support the development of guidelines for effective examination practices in the department.

Coordinator, Electronics Group, Electrical Engineering Dept. | Sep. 2021 – Aug. 2024
Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Coordination of the development and revision of academic programs, new courses/laboratories, and the acquisition of equipment for academic purposes.
- Coordinate the evaluation process of faculty and graduate student applicants.

Member, Faculty Search Committee | Sep. 2023 – Aug. 2024
Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation and interviewing of faculty applications, as well as lecturer and graduate assistant applicants.

Member, Academic Committee | Sep. 2023 – Aug. 2024
Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Manage the modification, update, and restructuring of EE undergraduate and graduate programs.
- Supervise the introduction of new courses and programs.
- Assess the quality of all EE programs and propose enhancement plans.

Treasurer, IEEE Western Saudi Arabia EDS Chapter | Nov. 2019 – Aug. 2023

- Maintaining financial accounts and overseeing all fundraising efforts of the chapter.

Member, Graduate Admission Committee | Sep. 2015 – Jun. 2023
Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation of graduate applications and recommendations to the graduate program.
- Review and update graduate needs for the department and identify targets and directions.

Chair, IEEE student branch | Jan. 2011 – Jun. 2014
King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.
- Meeting and hosting academic visitors.
- Support of several technical and social activities in the electrical engineering department.

Vice-chair, IEEE student branch | Sep. 2009 – Dec. 2010
King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.

EDITORIAL AND REVIEWING ACTIVITIES

Associate Editor | Jun. 2020 – Present
Editorial duties in Frontiers in Electronics (JIF 2.1)

Guest Editor | Oct. 2018 – Jan. 2019
Journal of Sensors (JIF 2.1)
Co-edited a special issue in "Flexible and/or Stretchable Sensor Systems".

Journal Reviewer for: Int. J. Mech. Sci. (JIF 9.4), Bioresour. Technol. (JIF 9), Compos. Struct. (JIF 7.1), Appl. Therm. Eng. (JIF 6.9), Adv. Mater. Technol. (JIF 6.2), Meas. (JIF 5.6), IEEE Electron Device Lett. (JIF 4.5), Rev. Adv. Mater. Sci. (JIF 3.9), Energy Technol. (JIF 3.6), J. Mech. Behav. Biomed. Mater. (JIF 3.5), IEEE Trans. Electron Devices (JIF 3.2), Flex. Print. Electron. (JIF 3.2), IEEE Sens. Lett. (JIF 2.4), IEEE J. Flex. Electron. (CiteScore 5.2).

HONORS AND AWARDS

- Outstanding Alumni Award at International Level, Electrical and Electronics Engineers Association (AIEEUN), 60th Anniversary, National University of Colombia, 2022.
- *Second Place*: "Fractal-based Electrodynamic Screen", 50K Challenge Award, KFUPM, May 2022. (Team advisor)
- Excellence in Teaching Award, College of Applied Engineering, KFUPM, Period 2020-2021.
- *First Prize*: "In-ear Health Monitoring System", Hope Hackathon, Digital Health Track, Saudi Arabia, 2020. (Team advisor)
- *Awarded Second Prize* for "Novel Wafer Thinning Processing Based on XeF_2 Etching for Flexible Electronics", on the KAUST 1st Graduate Research Symposium, PSE division, May 4–5, 2011.
- *Best presentation Award* for "Carbon Nanotube based Micro-Sized Microbial Fuel Cell", at the 5th Graduates' Seminar at King Fahd University of Petroleum and Minerals, May 8, 2011.
- *KAUST Academic Excellence Award* (Fall 2010–Spring 2011).
- *KAUST Graduate Fellowship and Provost Award* 2009. (In recognition to students with outstanding scholastic achievement and personal accomplishment).
- *German Academic Exchange Service (DAAD) - Young Engineers Scholarship* 2007. (Awarded for excellent academic performance and German language skills).
- Honorary Mention, 2002 *Colombian Physics Olympiad*.
- Gold Medal (2002), Honorary Mention (2001), *Colombian Chemistry Olympiad*.

TECHNICAL SKILLS AND MEMBERSHIPS

Languages	Spanish (native), English (fluent), German (Intermediate), Arabic (Basic).
Micro/Nanofabrication Skills	Photolithography, Wet Etch, Reactive Ion Etching (RIE), Deep Reactive Ion Etching (DRIE), Atomic Layer Deposition (ALD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Sputtering, Evaporation, Annealing.
Instrumentation and Characterisation	Oscilloscopes, function generators, LCR meters. Temperature-measurement systems. Probe stations and I–V characterisation systems. Scanning electron microscopy (SEM) and thin-film characterisation equipment.
IT Skills	COMSOL, Coventor, L-Edit, MATLAB, LaTeX, C++, Visual Basic, VHDL, LabVIEW, Assembler.
Education tools	Blackboard, Blackboard Ultra, Poll Everywhere, Mentimeter, Gradescope.
Memberships	Member, IEEE; Member, IEEE Electron Device Society.